FARIA · GAIA

Tiny angles, and a map of the Galaxy

The European Space Agency built Gaia to measure how stars shift as Earth goes around the Sun. From those tiny angles came distances, and from the distances, a map.

THE ANGLE LADDER

Each rung is sixty times finer

1°	A pinky, held at arm’s length	one degree
1′	A dime, seen face-on, at about 200 feet	one arcminute · about 60 meters
1″	A dime at about 2.3 miles	one arcsecond · about 4 kilometers

That factor of sixty is the old Babylonian way of dividing a circle, the same heritage that gives us sixty minutes in an hour.

A dime, as seen from the Moon

For bright stars, Gaia measures angles of a few tens of microarcseconds. A microarcsecond is one millionth of an arcsecond. That is about the width of a dime as seen from the Moon.

An arcsecond already leaves the solar system

A star whose parallax is one arcsecond sits one parsec away, about 3.26 light-years — already far beyond Neptune. Gaia works thousands of times finer than that.

Why the angles matter

The smaller the shift, the farther the star. Measuring those shifts is how Gaia turns a sky full of lights into a map with depth.

Planned for five years. It watched for ten and a half.

2013

19 December

Launched. Gaia took up a post far beyond the Moon.

2014–19

July 2014 to June 2019

The planned five-year science mission.

2025

15 January

Science ended, about 10.5 years in, when the cold gas ran out.

2026

2 December, Data Release 4

The first 5.5 years of data. The final catalogue is still to come.

The observations are so many, and the work of turning them into a catalogue so heavy, that the final release — from the whole 10.5-year mission — is not expected before the end of 2030.

Nearly two billion stars.

Not every star in the Milky Way.

The Galaxy holds a few hundred billion stars. Many are too faint, or hidden by dust, for Gaia to see. What it did measure is still the largest precise census of the brighter stars ever made.

What no one expected

Gaia-Enceladus

In 2018, Gaia showed that a large part of the Milky Way's inner halo came from another galaxy that merged with ours about 10 billion years ago. Its stars still move together the wrong way.

A snail-shell wobble

That same year, Gaia found a spiral in the motions of nearby stars. The Galaxy's disk is still wobbling, like a pond after a stone hits it, probably after the Sagittarius dwarf galaxy passed through.

Sleeping black holes

Gaia BH3, announced in 2024, is a black hole about 33 times the Sun's mass, 1,926 light-years away. It was the third such hole Gaia found, after BH1 and BH2 in 2023.